METRO TICKET GENERATING SYSTEM

Comprehensive Full Stack Project Documentation (FSD & Final Report Format)

Date	**2 October 2026**
Team ID	
Project Name	**Metro Ticket Generating System in ServiceNow Using Service Catalog and Workflows**
Maximum Marks	**4 Marks**

1. INTRODUCTION

- **Project Title:** Metro Ticket Generating System in ServiceNow using Service Catalog and Workflows.
- Team Members & Roles:
  - [Your Name] – Lead Full Stack Developer & ServiceNow Scoped Application Architect (Responsible for custom tables, Script Includes, Flow Designer workflows, and QR payload generation).
  - [Team Member 2] – Service Portal & UI Designer (Responsible for Record Producers, service portal widgets, and operator workspace views).
  - [Team Member 3] – QA & Testing Lead (Responsible for test cases, UAT execution, and workflow validation).

2. PROJECT OVERVIEW

- **Purpose:** To replace slow, manual, cash-dependent metro station counters with a fully automated, self-service ServiceNow platform solution (x_metro_ticket) that instantly captures travel requests, computes dynamic zone-based ticket pricing, generates unique cryptographic QR-code passes, and streamlines station operator verifications.
- **Key Features:** Self-service Service Catalog Record Producer, dynamic server-side Script Include fare calculator, automated Flow Designer validation and task routing, unique cryptographic QR-code generation, and station operator Configurable Workspace ticket scanning.

3. ARCHITECTURE

- **Frontend (Service Portal & Workspace):** Built using ServiceNow Service Portal components, HTML/CSS/Client Scripts for portal catalog forms, custom QR ticket rendering widgets, and the Configurable Agent Workspace for station personnel.
- **Backend (Flow Designer & Script Includes):** Powered by ServiceNow Workflow Studio / Flow Designer engines and server-side JavaScript Script Includes (MetroFareCalculator) handling event-driven pricing, approvals, and QR generation logic.
- **Database (Scoped Tables):** Built on secure ServiceNow platform custom database tables (x_metro_ticket_booking, x_metro_ticket_station, x_metro_ticket_smart_card), storing relational commuter bookings, station master data, and RITM mappings.

4. SETUP INSTRUCTIONS

- **Prerequisites:** Active ServiceNow Developer Instance (Washington DC or later release), scoped application creator permissions, and admin access.
- Installation Steps:

1. Log in to your ServiceNow Developer Instance and open Studio.
2. Create or import the custom scoped application package (x_metro_ticket).
3. Configure the custom tables (x_metro_ticket_station and x_metro_ticket_booking) with appropriate reference and string fields.
4. Import station master data (Origin/Destination zones and fare multiplier rates).
5. Publish the Record Producer to the Service Catalog and assign roles via Scoped Access Control Lists (ACLs).

5. FOLDER STRUCTURE

- **Client / Portal Structure:** Contains Service Portal widgets (widget-metro-qr), catalog item variables, client scripts, and portal page layouts.

- **Server / Application Structure:** Organized under the x_metro_ticket scope containing Script Includes (MetroFareCalculator), Flow Designer flows, Business Rules, Access Controls (ACLs), and outbound email notification definitions.

6. RUNNING THE APPLICATION

- **Portal Execution:** Navigate to the ServiceNow Service Portal URL, open the Service Catalog, select “Metro Ticket Request”, fill in travel details, and submit.
- **Backend Validation & Monitoring:** Platform admins and station operators monitor incoming requests, RITMs, script executions, and fulfillment tasks directly via the ServiceNow UI or Configurable Workspace.

7. API DOCUMENTATION

- **Endpoint / Script Include API:** Exposed via scoped Scripted REST APIs or server-side script calls.
- **Example Method:** MetroFareCalculator.calculateFare(originSysId, destinationSysId, ticketClass) → Returns JSON response containing computed fare amount and zone distance metrics.

8. AUTHENTICATION

- Handled natively via ServiceNow platform user authentication sessions, role-based access control (RBAC), and custom scoped Access Control Lists (ACLs) separating commuter portal access from station operator workspace permissions.

9. USER INTERFACE & SCREENSHOTS

- Features clean portal catalog forms for ticket selection, live status trackers on user dashboards, rendered QR-code pass components, and operator workspace verification queues.

10. TESTING

- **Strategy & Tools:** Verified via ServiceNow Automated Test Framework (ATF), manual unit testing of Script Includes, end-to-end Flow Designer execution audits, and comprehensive User Acceptance Testing (UAT).

11. SCREENSHOTS OR DEMO

- **GitHub Repository & Project Demo Link:** https://github.com/chandrasekhar-DEV-CODER
- **Dataset / Master Data:** Internal ServiceNow Station Master Table records.

12. KNOWN ISSUES

- Outbound email delivery depends on the developer instance SMTP configuration (email sending properties must be explicitly enabled for live notifications).

13. FUTURE ENHANCEMENTS

- Integration with IoT physical turnstile hardware and smart card NFC readers for automated gate unlocking upon QR scan.
- Integration of mobile wallet payment gateways for instant top-ups.